# Introduction to Data Science

**Syllabus:**
All topics covered from **Session 1 to Session 11**.

**Guidelines:**

This mid-term assessment is a group project where groups will **submit** a Python notebook along with a slide deck by **29th Sep**. on **1st October**, the groups will deliver a presentation followed by a Q&A session to the faculty.

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| **Mode of Assessment** | **Group Presentation.** Each group also submits a Python notebook and a slide deck. |
| **Weightage** | 50% |
| **Group Formation** | Students form their own groups of 4–5. The programme office places any student without a group after Monday, 21 September 2026. |
| **Presentation Duration** | 10 minutes per group: 8 minutes presentation + 2 minutes Q&A. |
| **Syllabus** | Sessions 1–11. |
| **Tool** | Python only (Jupyter Notebook, VS Code or Google Colab). KNIME and other no-code tools are not accepted. |
| **Exam Date** | Session 12. (1 October 2026) |
| **Key Deadlines** | Group list: Mon, 21 Sep 2026 |
| **Evaluation & Feedback** | Within 1 week of the exam date. |

**2. Objective:**
Pick a real business problem. Use past data to build a machine learning model in Python. Then tell the business what it should do.You will work through the full pipeline: frame the business question, explore and clean the data, build and compare models, tune the best one, and explain the results in business terms. The goal is to show clear thinking, correct Python work, and the ability to explain a data-driven decision.

**3. Groups and Datasets**

- **Groups:** 4 to 5 students. You will receive the group list by ***Monday, 21 September 2026***.
- **Source:** Kaggle, UCI Machine Learning Repository, data.gov.in, or any other public source. The data must link to a real business problem.
- **Size:** At least 1,000 rows and 8 columns. Very small datasets give unstable models and weak results.
- **Not from class:** Do not use any dataset from the class demos or exercises.
- **Approval:** Send your dataset link and business question to the instructor by Friday, 25 September 2026. Feel free to check with the instructor before you decide.
- **Rule:** One dataset per group. Two groups cannot use the same dataset. No change of dataset after approval.

- You may choose any dataset from the provided list. However, ***each group must select a unique dataset within their respective section***. Please ensure that no two groups from the same section choose the same dataset.

## 4. **Project Workflow (all steps in Python)**

- **Business Understanding —** Who is the stakeholder? What decision will the model support? What does success look like? Is this a regression, classification or clustering problem?
- **Data Understanding —** Explore the data with pandas and charts (matplotlib / seaborn). Check data types, distributions, missing values, outliers, correlations and class balance.
- **Data Preparation —** Treat missing values, encode categories, scale numbers and create useful new features. Split the data into train and test sets before you fit anything. Use a scikit-learn Pipeline so that no information from the test set leaks into training.
- **Modelling —** Build at least 3 models taught in class and compare them. For example: Linear Regression, Ridge / Lasso, Logistic Regression, Decision Tree, Random Forest, KNN, SVM. Also compare with a simple baseline, such as always predicting the average value or the most common class. K-Means clustering (for example, to segment customers) can be added as an extra.
- **Model Selection & Tuning —** Pick the best model using cross-validation. Tune its settings (hyperparameters) with GridSearchCV or RandomizedSearchCV. Show the score before and after tuning.
- **Evaluation —** Use the right metrics. Regression: $R^2$, RMSE, MAE. Classification: confusion matrix, accuracy, precision, recall, F1, ROC-AUC. Show which features matter most. Explain what the numbers mean for the business.
- **Conclusions —** Give a clear recommendation and the next steps. List the limitations and assumptions honestly.

## 4. **Project Workflow (all steps in Python)**

- **Due:** 11:59 pm, 2 days before the exam date. One member uploads the files for the whole group.
- **Python Notebook (.ipynb):** Must run from top to bottom without errors (Restart Kernel & Run All). Use a markdown heading for each workflow step, with short notes that explain what you did and why. Also submit an HTML or PDF copy of the notebook with all outputs visible.
- **Dataset:** The data file (or a link to it), with a short data dictionary that explains each column.
- **Presentation Deck (PPT or PDF):** 10–12 slides in 4 parts:
  (i) Business Question & Data Understanding,
  (ii) Data Preparation,
  (iii) Modelling, Tuning & Results
  (iv) Conclusions, Way Forward & Limitations. ***Add one last slide that shows who did what in the group.***
- **File names:** *GroupNo_ProjectName, for example Group3_LoanDefault.ipynb.*

## 6. Presentation Day

- **Time:** 8 minutes to present, then 2 minutes of Q&A. You will be stopped at 10 minutes.
- **Focus:** Show results and decisions, not code line by line. Keep the notebook open in case you are asked about a step.

- **Every member:** Any member can be asked any question. Each person should understand the full project, not only their own part.
- **AI tools:** You may use AI tools such as ChatGPT/Claude for help with code. You must still be able to explain every step. Cite any outside code, AI tools or references you use.

## 7. Presentation Rubrics- Total 100 Marks

| Component | Criteria | Marks |
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| **A. Project Quality** | | **30** |
| **Business & Data Understanding** | Clear business question and stakeholder. Useful EDA charts, with findings written in plain words. | |
| **Data Preparation** | Correct handling of missing values, encoding, scaling and new features. Train-test split before fitting. No data leakage. | |
| **Modelling & Tuning** | At least 3 models compared with each other and with a baseline. Cross-validation used. Best model tuned. | |
| **Evaluation** | Right metrics for the problem. Results explained in business terms. | |
| **Conclusions & Limitations** | Clear recommendation, way forward, and honest limitations. | |
| **Notebook Quality** | Runs from top to bottom. Clean, well-commented and easy to follow. | |
| **B. Class Presentation** | | **50** |
| **Clarity & Storytelling** | Clear flow from the business question to the recommendation. | |
| **Depth of Insight** | Goes beyond numbers. Explains what the results mean and why. | |

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| **Visuals** | Clean, readable charts and slides. | |
| **Delivery & Time** | Confident delivery, finished within 10 minutes. | |
| **C. Q & A (marked for each student)** | | **20** |
| **Answers** | Questions on the approach (3), choice of algorithm (4), and results & limitations (3). | |
| **TOTAL** | | **100** |

Sections A and B: the same marks for every member of the group. Section C: marked for each student. Originality, reproducibility and clarity of thought carry heavy weight.